NEF ADJOINT DIVISORS: FRAMEWORK – GENERAL DIMENSION

1. Theorem and proof

Fix an integer $d \geq 2$. We use the following source inputs, denoted collectively by (I_d).

- For every projective klt $\mathbb{Q}$-pair (Y, B) of dimension $j < d$ and every nef $\mathbb{Q}$-Cartier divisor M, if $K_Y + B$ is pseudo-effective and, for some $a \in \mathbb{Q}_{>0}$,
$$(K_Y + B + aM) \cdot C > 0$$
for every integral curve through a sufficiently very general point, then K_Y+B+cM is big for every $c \in \mathbb{Q}_{>0}$.
- Every projective log canonical $\mathbb{Q}$-pair of dimension at most d with nef adjoint has semiample adjoint.
- Every projective $\mathbb{Q}$-factorial generalized klt $\mathbb{Q}$-pair of dimension at most d, with fixed nef $\mathbb{Q}$-Cartier b-divisor data and pseudo-effective adjoint, admits a terminating choice of minimal-model program with a nef endpoint. The program extracts no divisors, preserves $\mathbb{Q}$-factorial generalized klt models, and on a common resolution the initial adjoint equals the final adjoint plus an effective divisor exceptional over the final model.

The first input is (P_j) of [1, Section 3.1]. The last two are the companion inputs used in [1, Section 2].

Theorem. *Assume* (I_d). *Let* X *be a projective klt variety over* $\mathbb{C}$ *of dimension* d, *with* K_X *pseudo-effective. Let* D *be a nef* $\mathbb{Q}$*-Cartier divisor such that*
$$L := K_X + D \quad \textit{is nef}, \qquad \mathrm{n}(X, L) = d.$$
Then L *is big.*

Proof. Suppose that L is not big. Since L is nef, $L^d = 0$. After a small projective $\mathbb{Q}$-factorialization, we may assume that X is $\mathbb{Q}$-factorial. The pullbacks of K_X, D, L retain the stated properties. Every integral curve through a very general point of X has positive L-degree.

Adjoint reduction. The minimal-model and abundance inputs give $\kappa(K_X) \geq 0$. If $\kappa(K_X) > 0$, the positive-Iitaka-dimension reduction in [1, Section 3.3], using (P_j) for $j < d$, makes L big, a contradiction. Thus $\kappa(K_X) = 0$.

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The two-nef-model lemma of [1, Section 3.2] gives a birational contraction $X \dashrightarrow Y$ and $s_0 \in \mathbb{Q}_{>0}$ such that Y is projective, $\mathbb{Q}$-factorial and klt,

$$K_Y \sim_{\mathbb{Q}} 0, \qquad N \text{ is nef}, \qquad \mathrm{n}(Y, N) = d,$$

where N is the transform of D. On a common smooth resolution $X \xleftarrow{p} V \xrightarrow{q} Y$, there is an effective q-exceptional $\mathbb{Q}$-divisor F such that

$$p^*(K_X + s_0 D) = q^*(K_Y + s_0 N) + F.$$

The divisor N is not big. Indeed, if it were big, this equality would make $K_X + s_0 D$ big. For a rational number $0 < \lambda < \min\{1, 1/s_0\}$, the identity

$$L = \lambda(K_X + s_0 D) + (1 - \lambda)K_X + (1 - \lambda s_0)D$$

would then make L big, since the last two summands are pseudo-effective.

Take the global index-one cover $\nu : Y^{\mathrm{ind}} \to Y$ and a crepant $\mathbb{Q}$-factorial terminalization $h : W \to Y^{\mathrm{ind}}$, as in [1, Section 7]. Choose an integer $b > 0$ such that $M := b(\nu \circ h)^* N$ is Cartier. Then

$$W \text{ is projective, } \mathbb{Q}\text{-factorial and terminal}, \qquad K_W \sim 0,$$
$$M \text{ is nef and Cartier}, \qquad \mathrm{n}(W, M) = d, \qquad M^d = 0.$$

Full nef dimension is preserved on the common very general locus. Bigness is detected by the generically finite pullback $\nu \circ h$, so M is not big. It remains to exclude this terminal counterexample.

Setup.

- Fix an ample Cartier divisor A on W.
- Fix $\epsilon \in \mathbb{Q}_{>0}$ sufficiently small. For $n \geq 1$, put

$$a_n = ((M + A/n)^d)^{1/d}, \qquad r_n = \max\{1, \lfloor \epsilon/a_n + 1/2 \rfloor\}, \qquad P_n = r_n(M + A/n).$$

  Then P_n is ample and $\mathbb{Q}$-Cartier, $r_n \to \infty$, and $P_n^d \to \epsilon^d$, since $M^d = 0$. The index n is independent of the dimension d.

- Let $p_i : W \times W \to W$ be the projections. Put

$$Z = \mathbb{P}_{W \times W}\big(p_1^* \mathcal{O}_W(M) \oplus p_2^* \mathcal{O}_W(M)\big), \qquad \pi : Z \to W \times W.$$

  Let S_1, S_2 be the two summand sections, and set

$$U = \pi^{-1}(W_{\mathrm{reg}} \times W_{\mathrm{reg}}) \setminus (S_1 \cup S_2).$$

  Thus U is smooth and $\dim Z = 2d + 1$.

- Set $\xi = c_1(\mathcal{O}_Z(1))$. For an integer $s > 0$, put

$$R_{n,s} = (p_1 \circ \pi)^* P_n + (p_2 \circ \pi)^* P_n + s\xi.$$

  We use the convention $\pi_* \mathcal{O}_Z(a) = \mathrm{Sym}^a(p_1^* \mathcal{O}_W(M) \oplus p_2^* \mathcal{O}_W(M))$ for $a \geq 0$. Whenever $|kR_{n,s}|$ is used, $k > 0$ is an integer with kP_n Cartier.

- Choose ϵ sufficiently small that

$$(14\epsilon)^d < \tfrac{1}{4}, \qquad 80\epsilon < \tfrac{3}{4}.$$

  The integer s is chosen after ϵ and before n tends to infinity. The multiple k is chosen after n.

Jet separation. Let V be a variety, let B be a Cartier divisor, and let $z \in V_{\mathrm{reg}}$. For $j \geq 0$, $|B|$ separates j-jets at z if

$$\forall u \in \mathcal{O}_{V,z}(B) \quad \exists \sigma \in H^0(V, \mathcal{O}_V(B)) \quad \sigma_z - u \in \mathfrak{m}_z^{j+1} \mathcal{O}_{V,z}(B).$$

For the terminal counterexample above, we use the following two lemmas.

Lemma 1.1. *Assume* (I_d). *Let* W, M *and* $R_{n,s}$ *be as in Setup. There exists an integer* $s > 0$ *such that, for every* $n \gg 0$, $|kR_{n,s}|$ *separates* $(2d+2)k$*-jets at a very general point* $z \in U$, *for all sufficiently divisible positive integers* k.

This is the large-Seshadri-constant construction of [1, Section 5], followed by the jet conversion in [1, Section 6.7].

Lemma 1.2. *Assume* (I_d). *Let* W, M *and* $R_{n,s}$ *be as in Setup. For every integer* $s > 0$ *and every* $n \gg 0$, $|kR_{n,s}|$ *does not separate* $(2d+2)k$*-jets at a very general point* $z \in U$, *for any positive integer* k *with* kP_n *Cartier.*

The terminal application in [1, Section 6.7] proves this obstruction on a smooth resolution $w : \overline{W} \to W$. Here $K_{\overline{W}} \sim E$ for an effective w-exceptional divisor E, and an ample perturbation of $w^* P_n$ supplies the complete-intersection flag. The jet system transfers on the isomorphism locus. The Frobenius determinant has rank p^{2d} and its vanishing order satisfies the incompatible bounds

$$\frac{3}{4} p^{2d} \leq \mathrm{ord}(\sigma) \leq \big(80\epsilon + O(p^{-1})\big) p^{2d}$$

for arbitrarily large residual characteristics p. All complex data are fixed before reduction.

Fix s supplied by Lemma 1.1, and choose n sufficiently large for both lemmas. For a sufficiently divisible k, Lemma 1.1 gives separation at a very general point of U, whereas Lemma 1.2 forbids it. This excludes the terminal counterexample and completes the adjoint reduction. Hence L is big. □

Remark. The original variety X need not be smooth or have trivial canonical divisor; no irregularity assumption is imposed. The condition K_X pseudo-effective is retained from the source. The trivial-canonical condition occurs only on the auxiliary terminal model W. The two jet lemmas are unchanged in mechanism; the adjoint reduction precedes their application. No hypothesis $\kappa(D) = -\infty$ is needed. The full-nef-dimension target is bigness of $K_X + D$; this does not assert nonvanishing for an arbitrary nef adjoint of smaller nef dimension.

Draft status. This is a conditional framework following [1, Sections 3–7]. The source inputs (I_d) and the inherited geometric and Frobenius arguments have not been independently verified here. The two jet lemmas record the source proof obligations; they are not independent proofs supplied by this framework.

References

[1] *Numerical Semiampleness of Nef Adjoint Divisors*, OpenAI mathematics collection, October 3, 2026, Sections 3–7. Source manuscript.